Week 20 — Extended reading

Critic-free RL, RLVR, and process reward models

STUDY-RL

May 21, 2026

How to use this list

Five core papers, ranked roughly by how often the rest of the field cites them. Each entry includes a one-paragraph summary, a *why this matters for week 20*, and an annotated key passage with two to three sentences of commentary. Sourced from `research/llm_rl_sota.md` (sections 3–5) and `research/safety_rl_sota.md` (section 5).

1 Shao et al. — DeepSeekMath / GRPO

Citation. Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., . . . , Wu, Y. (2024). *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models.* arXiv:2402.03300.

Summary. The DeepSeekMath paper introduces Group Relative Policy Optimisation (GRPO). It samples G completions per prompt, scores each with a reward, and forms the advantage by group-mean subtraction and group-standard-deviation division. The advantage is broadcast to every token of the completion and plugged into a PPO-style clipped surrogate with a KL-to-reference penalty — crucially *without* a value head. On the 7B DeepSeekMath-Base, GRPO with rule rewards reaches a 51.7% top-1 on MATH, beating prior open-weights baselines and matching some closed models.

Why it matters for this week. GRPO is now the dominant critic-free algorithm for LLM RL — every major 2025 reasoning-RL recipe (DeepSeek-R1, Kimi K1.5, Open-Reasoner-Zero, DAPO) builds on its loss formulation. The paper is also the cleanest derivation of the group-baseline-as-control-variate idea applied to language models.

> *"The reason we choose to use such an objective function is that the value function used in PPO is typically another model of comparable size as the policy model, which brings substantial memory and computational burden. . . GRPO, instead, estimates the baseline from group scores, significantly reducing training resources."*

This is the central thesis: the critic was paying twice — once in memory, once in fragility of the on-policy value fit — and the group mean is a strictly better baseline for end-of-sequence rewards because it is unbiased by construction and uses no learned parameters.

2 Ahmadian et al. — RLOO

Citation. Ahmadian, A., Cremer, C., Gallé, M., Fadaee, M., Kreutzer, J., Pietquin, O., . . . , Üstün, A. (2024). *Back to Basics: Revisiting REINFORCE Style Optimization for Learning from Human Feedback in LLMs.* arXiv:2402.14740.

Summary. The paper argues that PPO is overkill for RLHF: token-level advantages are unnecessary when the only reward signal is at the end of the sequence, and the value head

adds compute without adding signal. Their proposal is REINFORCE-Leave-One-Out: sample k completions per prompt, use the average of the other $k-1$ as the baseline for each, treat the whole completion as a single action, and take a sequence-level policy gradient step. On a 6.9B model, RLOO outperforms a heavily tuned PPO at 3× the throughput and 70% less GPU RAM.

Why it matters for this week. RLOO is the conceptually cleanest critic-free algorithm and the foundation on which GRPO sits. It is also the reference point for variance-reduction analysis in lectures, because its unbiasedness is exactly the LOO independence argument (Theorem **??** in `notes.tex`).

> *"Treating PPO's clipping and advantage estimation as* required *features... papers over the fact that, in the RLHF setting, the variance of the simple REINFORCE estimator with a leave-one-out baseline is small enough that the additional complexity of GAE, value-network fitting, and per-token credit assignment is not warranted."*

The passage is also a methodological lesson: it is easy in deep RL to inherit machinery from prior work without checking whether the assumptions it was solving for still apply. The PPO clip and GAE are brilliant in continuous-control settings with dense reward and long horizons; for terminal-reward, single-action LLM RL they are largely ornamental.

3 Liu et al. — Dr.GRPO

Citation. Liu, Z., Chen, C., Li, W., Qi, P., Pang, T., Du, C., Lee, W. S., Lin, M. (2025). *Understanding R1-Zero-Like Training: A Critical Perspective.* arXiv:2503.20783.

Summary. A careful audit of the original GRPO loss isolates two biases: the $1/|y|$ token average penalises long correct chains-of-thought relative to short accidental ones (the "length bias"), and the $1/(\sigma_r + \varepsilon)$ amplifies gradients on prompts whose group has small intra-variance (the "difficulty bias"). The Dr. GRPO fix is to sum rather than average over tokens, and to drop the standard-deviation divisor. The patched loss closes the length-vs-quality decoupling that many RLVR runs were silently exploiting.

Why it matters for this week. Dr. GRPO is the example par excellence of a debugging insight that a reader who only sees the algorithm via its formula will miss. The two changes are tiny — a denominator and a sum — but they shift the implicit regulariser of the loss in ways visible only when you plot completion-length distributions over training.

> *"The standard GRPO objective implicitly defines a non-uniform weighting over tokens and groups that bears no relation to the reward landscape; correcting it is a one-line change and uniformly improves both accuracy and chain-of-thought length stability."*

The deeper moral: an unbiased estimator at the *population* level can still be biased at the *loss surface* level if the per-example normaliser depends on quantities that correlate with the target. This echoes the classical statistical lesson on weighting under heteroskedastic noise.

4 DeepSeek-AI — DeepSeek-R1

Citation. DeepSeek-AI (2025). *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning.* arXiv:2501.12948.

Summary. R1 is the production-scale instantiation of GRPO + RLVR. The R1-Zero variant trains from a base (non-SFT) model purely on rule-rewarded math, code, and logic tasks, with no preference data and no demonstrations. Despite the cold start, the model develops a long chain-of-thought style spontaneously — the paper documents "aha moments" where the model verifies its own derivations mid-stream. R1 proper adds a brief cold-start SFT, then multiple rounds of RL, beating GPT-4-level baselines on AIME and MATH while remaining open-weights.

Why it matters for this week. R1 is the empirical demonstration that the GRPO + verifiable-reward recipe scales. It also empirically contradicts the conventional wisdom that one cannot do useful RL from a base model: with a strong base, the rule reward alone provides enough signal for emergent CoT.

> *"DeepSeek-R1-Zero showcases capabilities such as self-verification, reflection, and generating long CoTs... without any supervised data, indicating that reasoning capability can emerge purely through RL, provided the reward is verifiable."*

This is a strong claim. It rests on the base model already having latent reasoning circuits that RL surfaces, rather than RL creating them from scratch — consistent with the weak-to-strong narrative in week 21. The R1 verifier is a regex on the boxed final answer plus a code sandbox: a strikingly minimal interface for such a strong result.

5 Lightman et al. — PRM800K

Citation. Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., Cobbe, K. (2023). *Let's Verify Step by Step.* arXiv:2305.20050.

Summary. The paper releases PRM800K — 800,000 human step-labels on MATH problem solutions — and trains a process reward model that scores every intermediate step. At best-of-N inference, the PRM re-ranker pushes a 6B base solver from 52% to 78% on MATH at $N = 1860$, comfortably above the same model's ORM counterpart. The paper also studies which step features the human labellers attended to, and provides an early discussion of *distributional shift*: PRMs trained on human-written derivations transfer to model-written derivations only when the model's CoT distribution is close to the labelled corpus.

Why it matters for this week. PRM800K is the canonical PRM dataset and the empirical anchor for the PRM-vs-ORM debate in section 6 of `notes.tex`. It also pre-figures the auto-labelling race that produced Math-Shepherd and OmegaPRM by quietly noting that 800K human step labels cost a small fortune.

> *"Process supervision provides a more precise reward signal than outcome supervision and reduces the model's reliance on lucky end-state matches. The benefit grows with the complexity of the reasoning chain and the proportion of solutions in the training set that contain partially correct work."*

The last clause is the crux: PRM advantage scales with the share of "almost right" chains. If your model is right or comprehensively wrong, ORMs suffice. If most of its mistakes are last-step arithmetic on otherwise correct derivations, a PRM is a much sharper teaching signal.

Sourced from

- `research/llm_rl_sota.md` sections 3 (critic-free PG), 4 (RLVR), 5 (PRMs).
- `research/safety_rl_sota.md` section 5 (reward-hacking failure modes feeding into week 21).